AC Circuit Analysis



Circuit with Resistors

- We know resistors don't change the phase of the voltage or current.
- We can just use these equations to work out the values in a resistive circuit
  - Voltage = Current * Resistance
  - Power = Voltage * Current



V = I * R
120 = I * 100
I = 1.2A

P = V * I
P = 120 * 1.2
P = 144W

Circuits with Inductors

- For circuits with inductors, we have a phase shift
- When we have just an inductor in a circuit, we know the impedance of the circuit must equal the reactance of that inductor
- This means we need to change our equations:
- **Impedance = Inductive Reactance = *2π* Frequency * Inductance***
- Voltage = Current * Impedance



$$X_L = 2\pi f L$$
$$X_L = 2\pi * 1000 * 5$$
$$X_L = 31415.92654$$

$$V = IX_L$$
$$120 = I * 3141592654$$
$$I = 0.003819718634$$

Circuits with Capacitors



- For circuits with capacitors, we again have a phase shift
- When we have just a capacitor in a circuit, we know the impedance of the circuit must equal the reactance of that capacitor
- This means we need to use the equations:
- **Impedance = Capacitive Reactance =** $\frac{1}{2\pi * frequency * capacitance}$
- Voltage = Current * Impedance

$$X_C = \frac{1}{2\pi f C}$$

$$X_C = \frac{1}{2\pi * 1000 * 5}$$

$$X_C = 3.18309886 * 10^{-5}$$

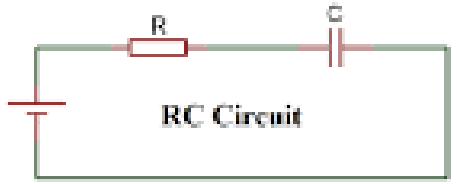
$$V = IX_C$$

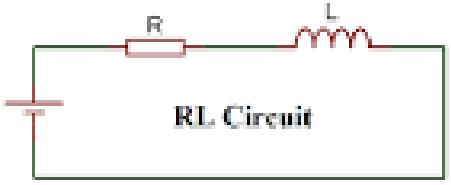
$$120 = I * 3.18309886 * 10^{-5}$$

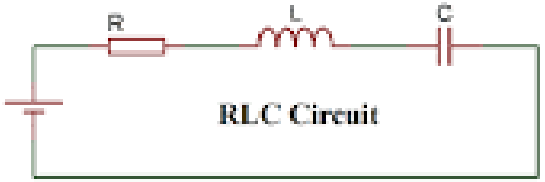
$$I = 3769911.184$$

RLC, RC and RL Circuits

- RLC Circuits contain all 3 components;
  - A Resistor (R)
  - An Inductor (L)
  - A Capacitor (C)
- RL Circuits contain two of the components;
  - A Resistor (R)
  - An Inductor (L)
- RC Circuits contain two of the components;
  - A Resistor (R)
  - A Capacitor (C)





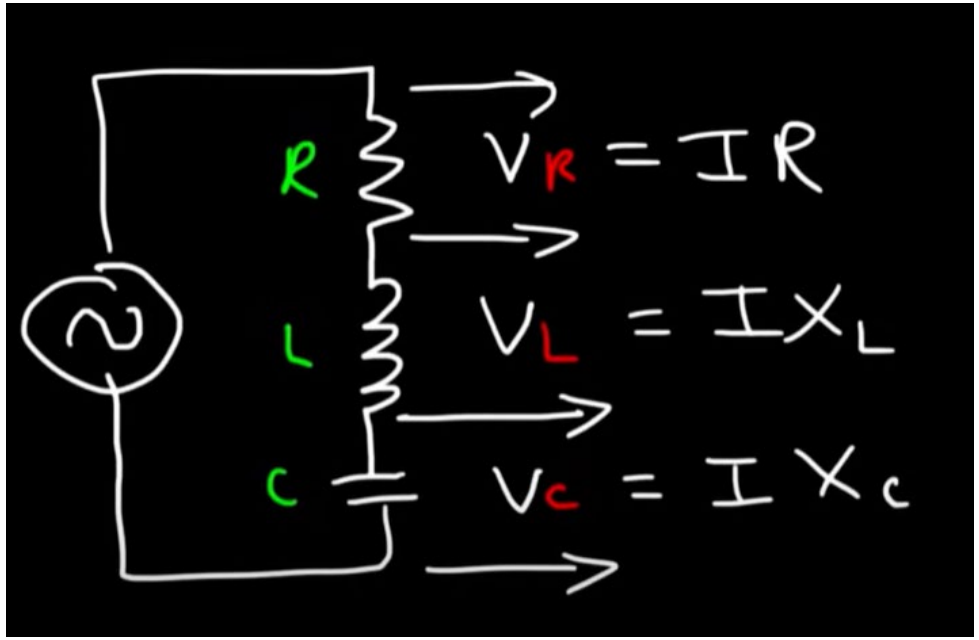


Simple Series RL, RC and RLC Circuits

- For each of the different series circuits you can use impedance to work out voltage and current using the equation:
  - $Z = \sqrt{R^2 + (X_L - X_C)^2}$ where:
    - Z = Impedance
    - R = Resistance
    - X_L = Inductive Reactance
    - X_C = Capacitive Reactance
    - Note when you don't have an inductor or a capacitator you just set them to 0

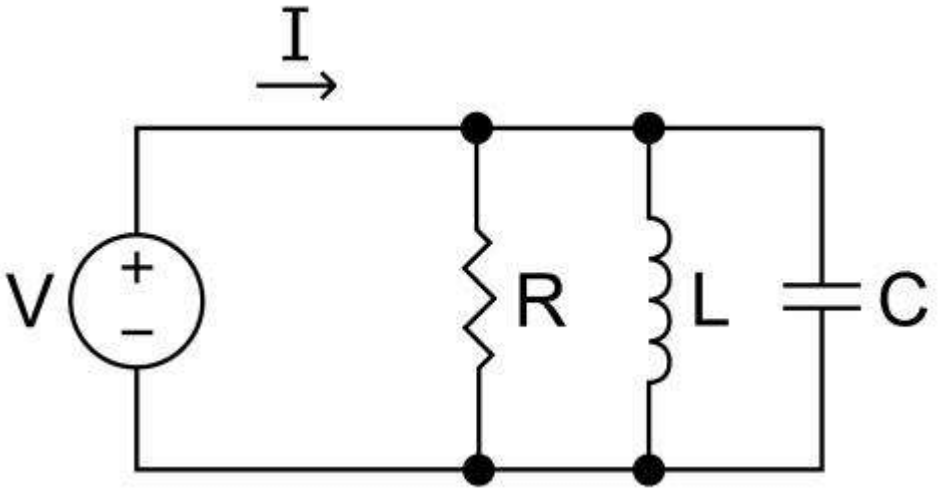
Series RL, RC and RLC Circuits

- We can then work out Voltage or Current for the whole circuit using the equation
  - $V = IZ$
- However, for each component in the circuit we use their individual equation



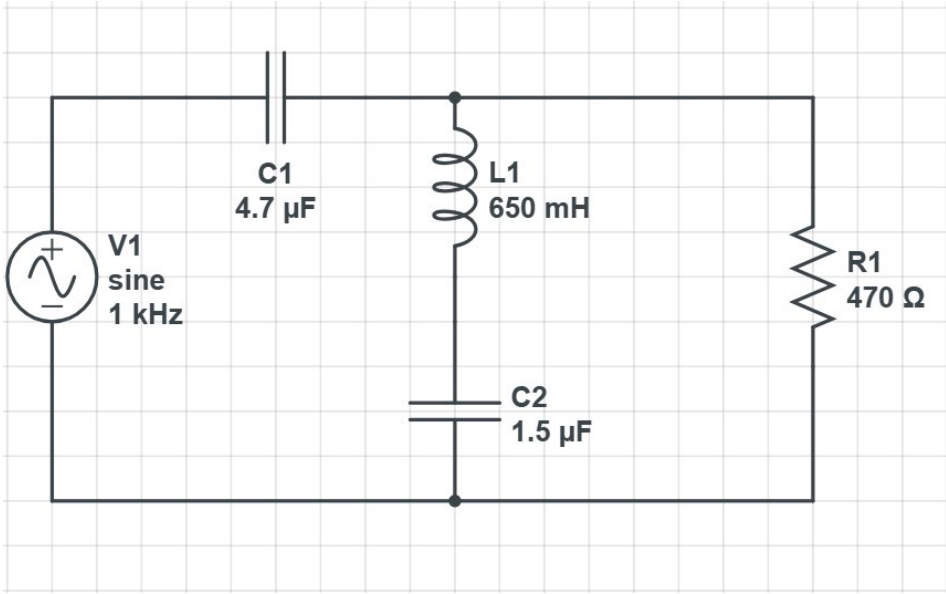
Simple Parallel RLC Circuits

- When we have components in parallel, we work out our impedance differently
- For standard single component circuits:
  - We use the equation: $Z = \frac{1}{\sqrt{\frac{1}{R}^2 + (\frac{1}{X_L} - \frac{1}{X_C})^2}}$
- Remembering we can sub in 0 if we are missing a capacitor or an inductor



Complex RLC Circuits

- Complex RLC circuits are circuits in which multiple different components make a more advanced circuit rather than just single components in parallel or series
- In general, I'd recommend always working in rectangular when doing complex RLC analysis
- Though you will have to convert to polar to find current and voltage values



Complex Series RLC Circuits

- When we have components in series in a complex circuit, we just add their resistance values together and then their reactance values together
- $(R_1 + i_1) + (R_2 + i_2) = \big((R_1 + R_2) + (i_1 + i_2)\big)$
- So for example if we have:
- $(120 + 320i) + (50 + 480i)$
- We get:
- $(120 + 320i) + (50 + 480i) = (170 + 800i)$

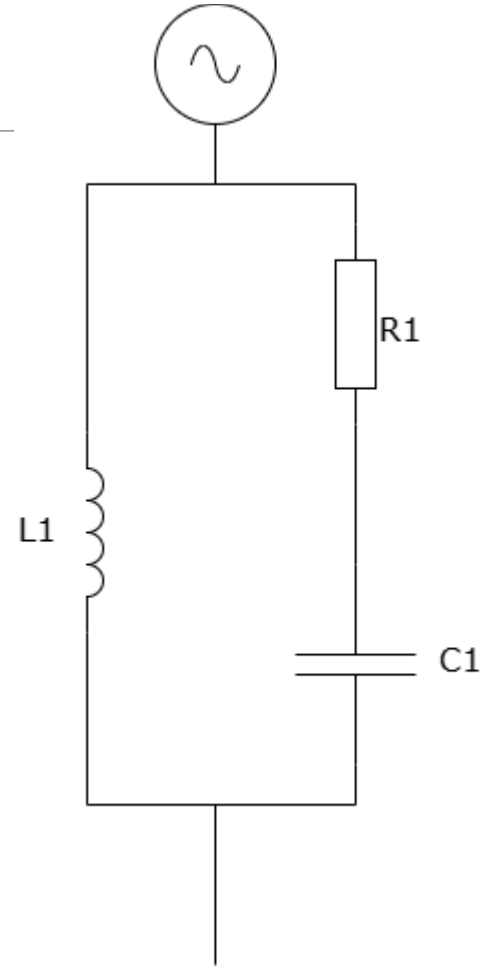
Note these are likely a combined components as they both have R and i values

Complex Parallel RLC Circuits

- For more complex RLC circuits we need to use $\frac{1}{\frac{1}{(R_1+i_1)}+\frac{1}{(R_2+i_2)}}$
- For this we need to use admittance $\left(\frac{1}{(R+i)}\right)$
- To work out the admittance we use: $\frac{1}{(R+i)} = \frac{R-i}{R^2+i^2}$

Remember to flip the sign here

- So for instance $(560 + 200i)$ becomes:
- $\frac{1}{(560+200i)} = \frac{560-200i}{560^2+200^2} = \frac{560-200i}{353600} = 0.001584 - 0.0005656i$

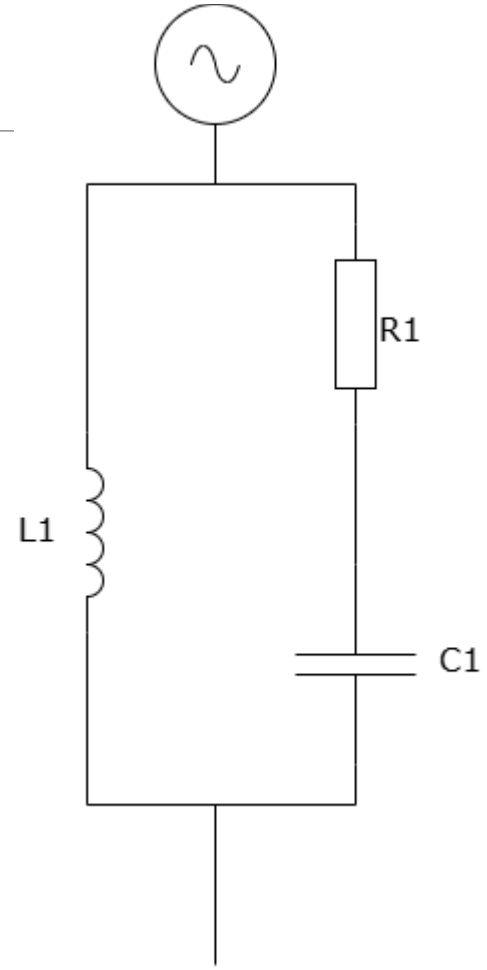


Complex Parallel RLC Circuits

- So for this example: $(120 + 320i)||(50 + 480i)$

Note the || just means in parallel with

- We do $\frac{1}{\frac{1}{120+320i}+\frac{1}{50+480i}}$
- So first we work out the admittance of each of the parts:
- $\frac{1}{120+320i} = \frac{120-320i}{120^2+320^2} = \frac{120-320i}{116800} = 0.001027 - 0.002740i$
- $\frac{1}{50+480i} = \frac{50-480i}{50^2+480^2} = \frac{50-480i}{232900} = 0.0002147 - 0.002061i$



Complex Parallel RLC Circuits

- Now we add the two admittances together
- $(0.001027 - 0.002740i) + (0.0002147 - 0.002061i)$

$$(0.001242 - 0.004801i)$$

- Then we convert back to impedance
- $\frac{1}{(0.001242-0.004801i)} = \frac{0.001242+0.004801i}{0.001242^2+0.004801^2} = \frac{0.001242+0.004801i}{0.00002459}$
- $(50.5039 + 195.2248i)$

